

12 **(short)** A straight infinitely long wire is placed in the plane of a square coil and parallel to one side as shown in Fig. 7. Each side of the coil is 10 mm long. What is the magnetic flux through the coil if its centre is 150 mm from the wire which carries a current of 2 A? If the current in the wire alternates at 50 Hz, approximately how many turns should the coil have to generate an alternating potential of $2\text{ }\mu\text{V}$ across its terminals?

[10]

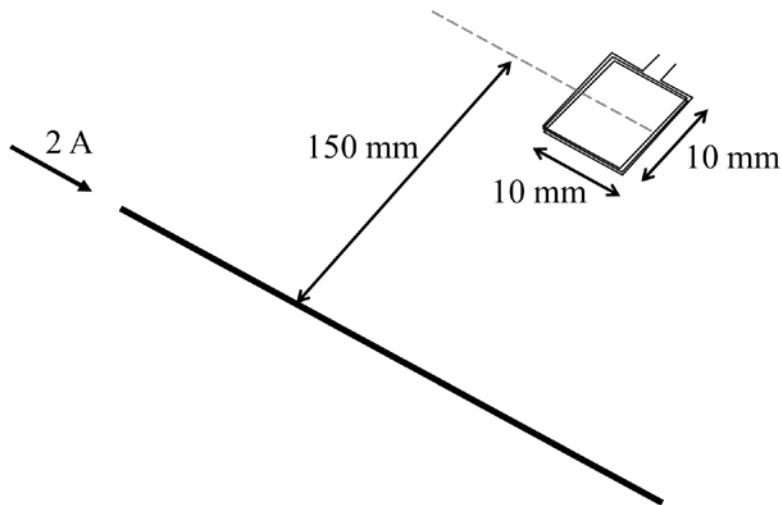


Fig. 7

END OF PAPER

NUMERICAL ANSWERS

Paper 3

ELECTRICAL & INFORMATION ENGINEERING

- 1 (b) $20\ \Omega$, $3.125\ \text{mW}$
- 2 (a) $489\ \text{nF}$
- (b) $14.3\ \text{A} \angle -59.6^\circ$
- 3 (a) $30 + j20\ \Omega$
- (b) $48\ \text{A}$
- (c) $691.2\ \text{W}$, $460.8\ \text{VAR}$
- (d) 75%
- 4 (c) $10\ \text{M}\Omega$, 0.976 , 97Ω
- (d) -0.005
- (e) $4.04\ \mu\text{F}$
- 5 (c) $5\ \text{k}\Omega$
- 7 (a) $-4.6875\ \text{V}$
- 10 (b) $1.72\ \text{A}$
- 12 $2.66 \times 10^{-10}\ \text{Wb}$, 24 turns